

Requirements and prototyping

IT Product Development

Innopolis University, 2026

Agenda

- Why this week
- The chain: vision, stories, prototype
- How to decide, not how to do
- Vocabulary
- Validating with the customer

Why this week

Last week

You settled:

- Which products exist, and how well each one does the job
- Where they all fall short: GAP-01, GAP-02
- What you will do differently: VP-01, VP-02
- What the customer disagrees with

You did not settle:

- What your product actually has to do
- What it will not do
- What you are building first

This week: say what it must do

The research said **what to build**.

This week you say **what it must do**, and you find out how much of that was wrong.

Conjectures lead to experiments. Experiments lead to facts. Facts lead to decisions.

Quiz

- What did last week's research give you that you can build on?
- Name one thing it did not give you.

The chain

The chain

GAP-nn, VP-nn -> the vision -> **US-nn** -> the prototype -> the customer -> **a change**

Every link answers one question:

- The vision: what is this for?
- A user story: what does one user need?
- Acceptance criteria: how would we know?
- The prototype: are we guessing?
- The meeting: where are we wrong?

The product vision

docs/product-vision.md, one page, five things:

- The **goal**: what the product must achieve
- The **constraints**: what you cannot change, and what each one costs
- The **stakeholders**: who uses it, who operates it, who is affected
- The **boundary**: what it will not do
- A **system context diagram**

The goal traces to a VP-nn. A goal that supports no value proposition is a feature you would like.

The boundary

The boundary is the list of things your product will **not** do.

It is what makes the context diagram checkable. With it, a reviewer can point at an actor and ask which of your exclusions rules them out. Without it, a diagram can only be judged as looking reasonable, and “reasonable” is not a test.

Your Won't Have stories are the same list, from the other direction.

A user story

As a <user>
I want to <action>
so that <value>

It is a **need**, not a design. “Add a redaction rule editor” is a feature you have already built in your head. “Attach a rule to a region of our own code” is a need you can still be wrong about.

The third line is the one that catches the difference. If you cannot finish it with a value, you have written a task.

Acceptance criteria

The story says **why**. The criteria say **what success looks like**.

Two or more per story, and the test is simple: could somebody who is not you run the check and get the same answer?

“Works well” is not a criterion. A criterion you cannot run is a belief.

A hard criterion is not a writing problem. It means you do not yet know what the product does in that case, which makes it a good thing to prototype.

The prototype

A prototype is an instrument for finding out what is wrong.

It is not a showcase, and it is not the product.

Ask it one question first: **which story am I least sure about?** Then build the cheapest thing that gets a reaction to that question.

Which user stories does the prototype cover? Answer that explicitly, out loud, in the file you record it in.

Quiz

- What question does each artifact in the chain answer?
- What is the difference between a need and a design?
- How do you tell a real acceptance criterion from a wish?

Two traps worth naming

Trap 1: a solution wearing a story's clothes. The story names a screen, a button, or a component. You have already designed it, so you can no longer be surprised by it.

Trap 2: a goal nobody can fail. “Be fast”, “be reliable”, “delight the user”. Nobody can tell on Friday whether you achieved it, so it cannot guide a decision on Wednesday.

One story, twice

Both are about a study group planner. Both are made up.

Not a story:

| As a student, I want a calendar page so that I can see all my study sessions in one place.

A story:

| As a student preparing for an exam, I want to see the sessions I have already missed, so that I can decide which one to reschedule before the week is over.

The first one is a feature with a screen in it. The second one has a situation, an object, and a decision.

Quiz

- True or false?: a story may name the screen it needs.
- What makes a goal falsifiable?
- Which of the two stories could surprise you?

Key Takeaways

Key takeaways: Write needs, not designs, or you have stopped learning. Make the goal something you could fail. The three words that give a story away are “page”, “button”, and “screen”.

Vocabulary

Six words

Goal: - what the product must achieve

Boundary: - what the team will not do

Constraint: - a condition you cannot change

Assumption: - a belief you have not verified

Story: - one thing a user needs

Criterion: - how you would know you delivered it

An assumption is not a constraint. A constraint you cannot change; an assumption you can be wrong about, and that is the interesting one.

Two distinctions that will cost you

A goal (Week 2) is not a quality goal (Week 4). A goal says **what**: the request leaves the network governed by the company's own rules. A quality goal says **how well**, and it arrives in Week 4 with its own Q-nn identifiers.

A threshold of success (Week 3) is not a goal restated. A goal is the outcome. A threshold is the measurable bar for calling it done. If you can paste your goals into Week 3 and change nothing, they are not yet goals.

Prototype, proof-of-concept, MUP, product

Proof-of-concept: - does this work at all technically

Prototype: - does this deliver the value, and it normally gets thrown away

Minimum usable product: - real code that delivers value to somebody

The product: - what you ship, weeks later

A prototype is not a MUP. If it is going to be code, it goes through planning like everything else.

Warning! Be suspicious of a spike that nobody wants to throw away. Engineers are fascinated with the machine, and a spike is not a small feature.

Quiz

- Which of the six words can be wrong, and how would you find out?
- Give a goal and a quality goal for the same product.
- What is the difference between a prototype and a MUP?

Key Takeaways

Key takeaways: A goal is an outcome, a quality goal is a measure, and a threshold is the bar. Three different things, three different weeks. Know which one you are writing, because the vocabulary does not forgive mixing them.

Validating with the customer

What the second meeting asks

You are not asking permission. You are finding out where you are wrong.

The target is one sentence: **which of these eight stories is wrong?**

Not the business goals. Those were settled last week. A later meeting settles one thing or two, and everything else is a conversation you do not need to have again.

Do not treat agreement as a result

“That sounds great” about your own idea has told you almost nothing.

If the customer agrees with everything, you showed them the answer rather than the question. The Disagreements table in your meeting report should not be empty.

Warning! A prototype that validated everything proved nothing, because you chose the parts you were already sure about.

Trace what changed

Something has to change as a result. In all four places:

- `prototypes.md` - what you showed, what they said
- `meeting-report.md` - which US-nn changed
- the story file - what it says now, dated
- the week report - what to look at first

A decision in the meeting report that never reaches the story file has not changed anything. That is the step that gets skipped.

Quiz

- What is the target of this meeting, in one sentence?
- What do you do when the customer approves everything?
- Where does a change have to be recorded?

Key Takeaways

Key takeaways: The meeting is a test, so it is allowed to fail. Agreement is the one result that means you asked the wrong question. And a change that only exists in the meeting report is not a change.

Review questions

- What does the chain give you that a list of features does not?
- Why did planning move to Week 3?
- What is the boundary for, and what does it make checkable?
- How do you tell a real acceptance criterion from a wish?
- Which of your stories would you prototype first, and why that one?
- What is the difference between a goal, a quality goal, and a threshold of success?

Questions?

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